An Empirical Study of Large Language Model Output Stability under Prompt Perturbation: Disentangling Perturbation Effects from Sampling Noise and Their Relationship to Task Correctness

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Abstract

Large language models often produce different responses to prompts that are semantically equivalent but worded differently, a phenomenon commonly referred to as prompt sensitivity. However, because these models also generate non-deterministic output under standard sampling, an observed difference between two prompt versions may reflect either the wording change or the model's own sampling randomness, and most existing studies do not separate these two sources of variation. This study proposes an empirical framework that disentangles perturbation effects from sampling noise by establishing, for each prompt, a noise baseline derived from multiple generations under fixed decoding parameters. Five categories of everyday, non-adversarial prompt perturbations (paraphrasing, reordering, formatting changes, context injection, and surface noise) will be applied to benchmark items drawn from four task types: factual question answering, mathematical reasoning, code generation, and open-ended writing. Two complementary metrics, semantic similarity (measured with Sentence-BERT) and task correctness, will be used to examine whether the noise-corrected sensitivity ranking of perturbation types is consistent across task types, and whether semantic drift is associated with changes in correctness. The study aims to provide a reusable, cross-task methodology for quantifying prompt sensitivity that accounts for the model's own output variability.

1 Background and Motivation

In using large language models for everyday academic tasks, I repeatedly noticed a pattern: when I phrased the same question in different ways, by reordering clauses, choosing different wording, or adding a polite remark, the model's answer sometimes changed noticeably, even though the two phrasings appeared to express exactly the same intent to me. This observation is the starting point of the present study. In an ongoing project on algorithmic robustness, I have been using parameter sensitivity analysis, including tornado diagrams, to characterize how the performance of classical algorithms degrades as input conditions deviate from their ideal values, with the central question being how much a given input perturbation changes the system's output. Applying this analytical perspective to large language models, by treating the prompt as the input, the generated text as the output, and changes in wording or structure as perturbations, forms the basic framework of this study.

The sensitivity of large language models to prompt wording is a central, yet still insufficiently characterized, issue in assessing their reliability. Users typically assume that as long as the underlying intent of a question remains unchanged, the model's response should also remain largely consistent. Whether this assumption holds, and under what conditions, has not yet been examined through systematic quantitative research. The following section reviews the relevant literature from the past two years and identifies the gap that this study aims to address.

2 Literature Review

The fact that large language models are highly sensitive to prompt wording was first demonstrated systematically in adversarial robustness research. Zhu et al. introduced the PromptBench benchmark, which applies character-level, word-level, sentence-level, and semantic-level perturbations (including typographical errors, synonym substitution, and sentence reordering) across 8 tasks and 13 datasets, and found that such seemingly minor modifications can reduce downstream task accuracy by up to 33% [1]. This result established that prompt sensitivity is a real and substantial problem, but the perturbations were constructed adversarially: the researchers deliberately searched for the wordings most likely to degrade model performance, which differs from the kind of wording variation that ordinary users produce unintentionally in daily use.

Razavi et al. shifted the focus toward a setting closer to realistic usage. Building on TriviaQA and HotpotQA, they constructed semantically equivalent natural rephrasings rather than deliberately adversarial samples, and proposed the task of prompt sensitivity prediction: determining, without running the model, which rephrasings are likely to cause noticeable changes in answer quality [2]. They found that existing approaches, including model self-assessment and text classification methods, performed poorly on this prediction task, indicating that whether a given rephrasing will trigger a change in output remains difficult to anticipate and can currently only be observed empirically.

Chatterjee et al. proposed POSIX, a metric that converts sensitivity into a single computable value by measuring the relative change in the log-likelihood that a model assigns to the same response before and after an intent-preserving rephrasing [3]. This approach has the advantage of producing a single comparable score, but it requires access to the model's internal token probabilities and is therefore inapplicable to most commercial models that are only accessible through an API. Notably, their experiments also showed that for multiple-choice tasks, changes to the prompt template are the dominant source of sensitivity, whereas for open-ended generation tasks, paraphrasing is the dominant source. This finding suggests that which type of perturbation has the greatest impact on output may itself depend on task type, although POSIX was not used to compare multiple task types systematically within a single framework.

Agrawal et al. examined the problem at the level of task instructions. By applying character-level and word-level edits to the instructions of classification tasks (CoLA, QNLI, and SST-2), they found that such seemingly minor changes substantially reduce downstream performance, and proposed mitigation strategies such as self-denoising [5]. This work shows that prompt sensitivity arises not only in the wording of individual questions or examples, but also in the wording of task instructions, and that its magnitude is large enough to be of practical concern to developers.

A related quantitative approach was proposed by Errica et al., who defined two metrics for classification tasks: sensitivity, the entropy of the predicted class distribution across multiple semantically equivalent rephrasings of the same input, and consistency, the similarity of predicted distributions across different inputs belonging to the same class [4]. To eliminate interference from the randomness inherent in generation, they set the decoding temperature to zero so that model behavior is close to deterministic, allowing any observed change in predictions to be attributed to the rephrasing itself. While this design effectively isolates the effect of rephrasing, it also assumes that model output is approximately deterministic, and does not address how to separate the variation caused by rephrasing from the variation caused by sampling itself when the model generates at a non-zero temperature, which is the common setting for open-ended tasks such as creative writing or code generation.

This issue was quantified directly in a more recent study. Haase et al. performed a variance decomposition analysis across 12 large language models and 10 creative writing prompts, generating 100 outputs per combination (12,000 generations in total), and found that, depending on the specific metric, the proportion of variance explained by the model's own sampling randomness ranges from 10% to 34%, comparable to or even larger than the effect of the prompt itself. They argue explicitly that evaluations based on a single generation conflate sampling noise with genuine prompt or model effects [7]. This finding provides direct empirical motivation for the core methodological design of the present study, which establishes a noise baseline through repeated sampling and compares perturbation effects against it: without first subtracting sampling noise, an observed drift after perturbation could simply reflect sampling randomness rather than the perturbation itself.

Taken together, these strands of work point to a gap that has not yet been addressed. PromptBench [1] and the work of Agrawal et al. [5] show that perturbations at the prompt or instruction level can substantially affect model output, but the former focuses on adversarial perturbations and the latter is validated only on classification tasks. POSIX [3] suggests that the pattern of perturbation sensitivity may vary by task type, but does not provide a systematic comparison across task types. Errica et al. [4] propose a way to quantify sensitivity and consistency under deterministic decoding, but do not extend this to non-deterministic generation or to open-ended tasks. Haase et al. [7] demonstrate that the magnitude of sampling noise itself cannot be ignored, but their work concerns the overall decomposition of output variance attributable to the model and the prompt, rather than whether a specific type of perturbation produces a significant drift, and it does not address task correctness.

The present study combines a noise-baseline comparison, methodologically motivated by the finding of Haase et al. [7] that sampling noise is non-negligible, with a cross-task perturbation comparison framework that echoes the task-dependent sensitivity patterns reported in POSIX [3] and draws on the perturbation construction approaches of PromptBench [1] and Agrawal et al. [5]. By introducing two complementary metrics, semantic similarity and task correctness, this study examines whether the noise-corrected ranking of prompt sensitivity is consistent across four task types (factual question answering, mathematical reasoning, code generation, and open-ended writing), and whether semantic drift predicts changes in correctness. This integrates three previously separate lines of work, namely perturbation construction, task-dependent sensitivity patterns, and the quantification of sampling noise, into a single empirical framework.

3 Research Questions

Based on the literature review above, this study addresses the following three research questions.

RQ1. After applying a noise-baseline correction (that is, subtracting the random variation arising from the model's own repeated generations), is the ranking of the five perturbation types (paraphrasing, reordering, formatting changes, context injection, and surface noise) by their effect on output semantic similarity consistent across the four task types (factual question answering, mathematical reasoning, code generation, and open-ended writing), or is this ranking itself task-dependent?

RQ2. For tasks with a well-defined correctness criterion (factual question answering, mathematical reasoning, and code generation), is there a statistically significant association between the degree of semantic drift (the magnitude of the similarity decrease) and the probability that correctness changes (from correct to incorrect, or vice versa)? Does this association have the same strength across task types? Open-ended writing is excluded from this analysis because it lacks an objective correctness criterion; consequently, the conclusions of RQ2 apply only to the three tasks above, and whether they generalize to open-ended generation requires further study.

RQ3 (exploratory). Do the sensitivity ranking and the drift-correctness association identified above show a consistent pattern between an open-source model and a commercial API model, or are these patterns model-specific?

4 Methodology

4.1 Tasks and Benchmark Data

To avoid the selection bias that would result from constructing all items independently, and to ensure that results are comparable, benchmark prompts will be sampled from established public datasets: approximately 20 items for factual question answering will be drawn from TriviaQA, approximately 20 items for mathematical reasoning from GSM8K, and approximately 15 items for code generation from HumanEval, each accompanied by unit tests that allow functional correctness to be determined directly. Because no standard dataset exists for open-ended writing, approximately 10 open-ended questions without reference answers will be designed for this study; these will be used only for the semantic drift analysis and will not be included in the correctness analysis of RQ2.

4.2 Perturbation Design

Drawing on the perturbation taxonomies proposed in PromptBench [1] and by Agrawal et al. [5], and reflecting this study's focus on natural everyday rephrasing rather than adversarial attack, five perturbation types are defined: paraphrasing (meaning-preserving rewording), reordering (rearranging sentences or the order in which information is presented), formatting changes (such as converting text to a list or adding or removing polite phrases), context injection (adding background information unrelated to the question), and surface noise (simulating unintentional spelling or punctuation errors that occur in real user input, such as missing letters or extra spaces, rather than systematic adversarial attacks). One version of each of the five perturbation types will be generated for every benchmark prompt. All generated perturbed versions will be manually checked to confirm that they are semantically equivalent to the original prompt; versions that fail this check will be regenerated, to ensure that any subsequent difference in output reflects a change in wording rather than a change in the question itself.

4.3 Noise Baseline Design

This is the core of the proposed methodology. The work of Haase et al. shows that the variance contributed by the model's own sampling randomness can reach roughly thirty percent of total output variance, on the same order of magnitude as the effect of prompt rewording itself [7]. This implies that if each prompt version is used to generate only a single output, an observed drift after perturbation could simply reflect sampling noise. To rule out this confound, each original prompt and its five perturbed versions will be used to generate multiple outputs (3 to 5 are planned) under fixed decoding parameters (temperature set to 0.7 and top_p set to 0.9, held constant throughout the experiment). The distribution of semantic similarities among the multiple outputs generated from the same prompt constitutes that prompt's noise baseline. The similarity between the original prompt and each perturbed version will then be compared against this baseline, and a decrease in similarity will be attributed to a perturbation effect, rather than to sampling noise, only if it significantly exceeds the variation observed within the noise baseline.

4.4 Model Selection

The study plans to evaluate one locally deployed open-source model (such as Llama 3.1 8B run through Ollama) and one or two commercial API models (such as lower-cost options like GPT-4o-mini or Gemini Flash). The open-source model can be queried without limit, which facilitates the repeated sampling required for RQ1 and RQ2; the commercial models will be used for the cross-model comparison in RQ3.

4.5 Measurement Metrics

The degree of semantic drift in model output will be measured as the cosine similarity between the original and perturbed outputs, computed using Sentence-BERT embeddings [6]. Correctness will be determined differently across tasks: for factual question answering and mathematical reasoning, outputs will be compared against reference answers, while for code generation, correctness will be determined by running the unit tests provided with each HumanEval item.

4.6 Analysis Methods and Visualization

A repeated-measures analysis of variance, with item as the within-subjects factor, will be used to test whether perturbation type and task type have significant main effects and interaction effects on semantic similarity, followed by Tukey's HSD for pairwise post hoc comparisons. The difference between the post-perturbation similarity and the noise-baseline similarity will be assessed for significance using paired tests, and its magnitude will be quantified using an effect size such as Cohen's d. If time permits, a linear mixed-effects model, with item as a random intercept and perturbation type and task type as fixed effects, will be used as a robustness check to further address the non-independence inherent in repeated measurements. For visualization, three figures are planned: a bar chart ranking the five perturbation types by their average reduction in similarity, providing an overall summary; a heat map with perturbation type as rows and task type as columns, illustrating the interaction effect relevant to RQ1; and a box plot comparing the similarity distribution under each perturbation type with the corresponding noise-baseline distribution, to show directly whether the perturbation effect exceeds the range of random variation. The drift-correctness association addressed in RQ2 will be examined using logistic regression, with whether correctness changed as the dependent variable and the magnitude of the similarity decrease as the predictor, fitted separately for each task type and then compared across task types.

5 Expected Outcomes and Significance

The principal output of this study is a noise-corrected, cross-task comparison of prompt sensitivity. If RQ1 reveals significant differences in the sensitivity ranking across task types, this would be consistent with the task-dependent sensitivity patterns observed in POSIX [3] and would extend that finding by providing noise-corrected comparative evidence across four task types. If the ranking is instead relatively consistent across task types, this would suggest that certain perturbation types pose a broadly applicable risk regardless of task, which would itself be a noteworthy finding.

If RQ2 shows a significant association between the degree of semantic drift and changes in correctness, this would provide preliminary empirical evidence on whether semantic similarity can serve as an early indicator of a change in correctness, which would be relevant to applications where model outputs cannot be checked manually one by one, such as educational support tools.

In addition, this study translates the finding of Haase et al. [7] regarding the magnitude of sampling noise into a practical measurement procedure that combines a noise-baseline comparison, a cross-task comparison, and two complementary metrics (semantic drift and correctness). The code and procedure developed for this study can be reused directly by future researchers for other models or task combinations.

6 Limitations

The conclusions of this study should be interpreted in light of the following limitations. First, owing to constraints of time and resources, the number of benchmark items per task (15 to 20) and the number of repeated samples (3 to 5) are relatively small, so statistical power may be insufficient to detect small effect sizes; the results of this study should therefore be regarded as exploratory rather than conclusive. Second, model coverage is limited to one open-source model and one or two commercial API models, so the cross-model conclusions of RQ3 cannot be generalized to all large language models, particularly to larger frontier models, which may exhibit different patterns. Third, the boundaries between the five perturbation types are not entirely mutually exclusive (for example, paraphrasing and reordering can overlap in some cases); this taxonomy is adapted in part from existing work [1, 5], but will need to be refined and fixed through small-scale piloting before the main experiment. In addition, owing to resource constraints, only one version of each perturbation type is generated per item, so the effect of the perturbation type itself cannot be fully separated from the effect of the specific wording used to implement it; repeating this process across 15 to 20 items per task partially mitigates this issue, but the observed effect of each perturbation type should still be understood as the effect of that type as implemented in this study, rather than the average effect across all possible wordings of that type. Fourth, the semantic similarity computed using Sentence-BERT is an approximate measure of whether the meaning of a response has changed; it does not directly indicate whether a response is accurate or aligned with the user's actual intent, which is why this study also incorporates a correctness metric. However, because open-ended writing lacks an objective correctness criterion, the analysis of this task is limited to semantic drift. Finally, rate limits and usage quotas for commercial APIs may impose practical constraints during the experiment; if this occurs, the number of repeated samples or the number of models evaluated may need to be reduced accordingly, which would further reduce statistical power.

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